Ashok S Babu

Software Developer

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science Engineering student with hands-on experience in software development, backend engineering, and AI/IoT applications. Proficient in Java, Python, JavaScript, Data Structures and Algorithms, REST API development, and database technologies including MySQL, MongoDB, and Supabase. Experienced in building scalable backend services, integrating APIs, developing full-stack applications, and applying software engineering principles to real-world problems.

EDUCATION

Adamas University **2023 - 2027**
Bachelor of Technology, Computer Science & Engineering

EXPERIENCE

InAmigos Foundation **21 Jul 2026 – 06 Aug 2026**
AI Data Analyst Intern *Remote*

- Conducted AI-assisted research and data analysis on NGO trends, including fundraising, CSR spending, and AI adoption.
- Collected and verified data, prepared reports and spreadsheets, and provided practical recommendations based on the analysis.

PROJECTS

PASAFE: Intelligent Vehicle Safety and Alert System | *Python - Js - Node.js - Raspberry Pi - Supabase*

- Engineered an IoT-based vehicle safety system using Raspberry Pi, GPS, GSM and sensors for real-time accident detection and emergency response.
- Implemented JavaScript/Node.js backend APIs and Supabase workflows to process sensor data, synchronize device data and trigger automated emergency alerts.
- Optimized IoT–backend integration by debugging API communication, device connectivity and data synchronization issues for reliable real-time operation.

Exoplanet Detection & Classification | *Python - TensorFlow/Keras - Lightkurve - AstroPy*

- Implemented a physics-guided AI pipeline for detecting and classifying potential exoplanets from noisy Kepler and TESS stellar light curves using Box Least Squares (BLS) transit detection.
- Architected a multi-stage approach combining 1D CNN temporal and 2D CNN spatial feature extraction from Target Pixel Files (TPFs), with hybrid feature fusion for multi-class classification and false-positive filtering.
- Integrated an Explainable AI layer using SHAP and Grad-CAM to interpret predictions and highlight influential transit regions and pixels.

AI-Based Crop Disease Detection | *Python - Pandas - PyTorch - NumPy - Matplotlib*

- Engineered a deep-learning crop disease classification system using the PlantVillage dataset and PyTorch for multi-class disease detection.
- Optimized neural network performance by identifying overfitting through validation analysis and applying data augmentation, regularization and dropout to improve generalization.
- Automated disease diagnosis outputs with treatment recommendations to support data-driven and smart agriculture.

TECHNICAL SKILLS

Programming Languages: Java,Python,JavaScript
Frontend: HTML5,CSS,React.js,
Backend: Node.js,REST APIs
Databases: SQlite, MongoDB,Supabase, Neo4j
AI / Machine Learning: PyTorch, TensorFlow,NumPy, Pandas
Developer Tools: Git, GitHub,Vercel, Render

ACHIEVEMENTS

- 2nd Place Winner at IIT Bombay Techfest 2025 for PASAFE Intelligent Vehicle Safety System.